products_03-prayog_ORDERING

3 Prayog STEM lab kit: ordering guide (what to buy, where, in what order)

> Companion to MANUFACTURING_MANUAL.md. Main caveat for this project: sensors uncalibrated. Prices and stock are from LCSC on the run date; vendors, shipping and customs rules change: verify before paying. General rules and vendor list: manuals/ORDERING_GUIDE.md.

1. Order of work

1. Read the manual and the caveat above.
2. Order the circuit boards and SMD assembly (section 2).
3. While the boards travel, order the other items (sections 3 and 5) and make the printed or laser-cut parts.
4. Assemble, program and test only after everything has arrived; follow the manual.

2. Order the circuit boards (JLCPCB, PCB + assembly)

Files (all in this repository):

- Gerbers: `utility10/pcb/release/prayog/1_fabrication/prayog_hub_gerbers.zip`
- Assembly BOM: `utility10/pcb/release/prayog/2_assembly/prayog_hub_JLC_BOM.csv`
- Pick-and-place (CPL): `utility10/pcb/release/prayog/2_assembly/prayog_hub_JLC_CPL.csv`
- Full BOM with LCSC numbers and prices: `costing/prayog_priced_BOM.csv`

Steps:

1. Create an account at **jlcpcb.com** and click *Order now*. Upload the Gerber ZIP. The viewer must show the board outline, both copper layers, silkscreen and drill holes.
2. Settings: **2 layers, 1.6 mm, FR-4**, lead-free HASL (use **ENIG** if the board has an LGA/QFN fine-pitch part, e.g. ADXL345, ESP32 module pad, BQ25570), green mask, white silk, quantity 5 (the minimum), *Remove order number: specify a location* if you want to avoid the order-number mark. Heavier copper (2 oz) only where the board notes say so.
3. Switch on **PCB Assembly**: *Economic*, assembly side *Top*, and set tooling holes *Added by JLCPCB* unless the board already has them. Choose the number of boards to assemble (2 or 5).
4. Upload the BOM and CPL. In the parts step, every line must match (LCSC numbers are in the BOM). Where a part is shown as *out of stock* or *no match*, use the alternative suggested by JLCPCB only after checking footprint and value against the list below; do not accept a different footprint.
5. In the **component placement preview**, check pin 1 / polarity of every IC, diode, LED, electrolytic capacitor, USB-C and connector against `*_assembly_top.pdf` in the same folder. Wrong rotation is the most common assembly error: ask JLCPCB support to rotate before approving.
6. Pay, then approve the assembly engineering review messages (they usually ask about a part or rotation). Delivery by DHL/FedEx/UPS takes about 1 to 2 weeks to India.

Parts **not** assembled by JLCPCB (you buy and solder them yourself):

Refs	Value	Part	LCSC
J2	ISP	HCTL PZ254-2-03-Z-8.5 (ZHOURI C5116479 had only 340 in stock)	C2894964
J3	Analog socket A0	JST B3B-PH-K-S(LF)(SN)	C131339
J4	Analog socket A1	JST B3B-PH-K-S(LF)(SN)	C131339
J5	Analog socket A2	JST B3B-PH-K-S(LF)(SN)	C131339
J6	Analog socket A3	JST B3B-PH-K-S(LF)(SN)	C131339
J7	Analog socket A4	JST B3B-PH-K-S(LF)(SN)	C131339
J8	Analog socket A5	JST B3B-PH-K-S(LF)(SN)	C131339
J9	Analog socket A6	JST B3B-PH-K-S(LF)(SN)	C131339
J10	Analog socket A7	JST B3B-PH-K-S(LF)(SN)	C131339
J11	Photogate	JST B3B-PH-K-S(LF)(SN)	C131339
J12	Ultrasonic HC-SR04	JST B4B-PH-K-S(LF)(SN)	C131334
TP1, TP2	TP		

Order these with the same LCSC order (add them to the cart so one shipment arrives) or buy locally in India (section 5).

Parts cost and stock (live LCSC prices on the run date)

Total parts per board: **US4.17*for1board* * *, * * *US*3.77 each for 10 boards**, **US$3.21 each for 100 boards** (about INR 367 / 332 / 282 at 88 INR/USD). The per-line break-up is in `costing/prayog_priced_BOM.csv`. This is parts only: PCB fabrication, assembly fees, shipping, customs and GST come on top (see `manuals/ORDERING_GUIDE.md`, section 6).

Stock warnings on the run date:

- TP (no lcsc price (hand-fit or not priced), stock)

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
U1	1	ATmega328P-AU	ATMEGA328P-AU	C14877	2.3308	2.331
C1,C2,C3,C6,C7,C8	6	100n	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	0.0075	0.045
R7	1	10k	0603 10k 1%	C25804	0.0024	0.002
J2	1	ISP	HCTL PZ254-2-03-Z-8.5 (ZHOURI C5116479 had only 340 in stock)	C2894964	0.0766	0.077
J1	1	USB-C data + 5V	TYPE-C-31-M-12	C165948	0.1890	0.189
F1	1	PTC 500mA	Littelfuse 1206L050YR (6 V)	C163512	0.0742	0.074
D3	1	SMAJ5.0A	Littelfuse SMAJ5.0A	C83329	0.1153	0.115
R1,R2	2	5.1k	0603 5.1k 1%	C23186	0.0016	0.003
C5	1	10u	0805 10uF X5R	C15850	0.0441	0.044
U2	1	CH340C	CH340C	C84681	0.5509	0.551
Y1	1	16 MHz	YXC X322516MOB4SI 16 MHz 12 pF 3225 (LCSC listing seen in search results; matches the 22 pF load caps)	C12668	0.0982	0.098
C9,C10	2	22p	Samsung CL10C220JB8NNNC C0G	C1653	0.0090	0.018

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
J3	1	Analog socket A0	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J4	1	Analog socket A1	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J5	1	Analog socket A2	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J6	1	Analog socket A3	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J7	1	Analog socket A4	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J8	1	Analog socket A5	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J9	1	Analog socket A6	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J10	1	Analog socket A7	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J11	1	Photogate	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J12	1	Ultrasonic HC-SR04	JST B4B-PH-K-S(LF)(SN)	C131334	0.0458	0.046
SW2	1	Reaction pad	Omron B3U-1000P	C231329	0.1894	0.189
TP1,TP2	2	TP				
D1	1	Green	Hubei KENTO KT-0603YG yellow-green 570 nm, Vf about 2 V (lights well from 3.3 V with 1 k; the Everlight 19-217 green was out of stock)	C2289	0.0105	0.011
R5	1	1k	0603 1k 1%	C21190	0.0017	0.002

2b. Enclosure: printed parts and fasteners (generated from the board)

- Print `4_enclosure/` base and lid (outer about 115 x 87 x 21 mm) in PETG or ABS, 3 perimeters, 25 % infill; the lid closes with **4 x M2.5 x 10 mm self-tapping screws** into the corner columns; the board sits on its standoffs with 4 x M2.5 x 6 mm self-tapping screws.
- Print `*_enclosure_caps.stl` (light pipe, plunger): plunger for SW2; light pipe for D1. Light pipes in clear PETG or resin; fit plungers and pipes from inside before closing the lid.
- indoor, dry use; print in PETG or ABS, 3 perimeters, 25 % infill
- Print one set first and test-fit the board, the holder and every cap; the dimensions are first-pass.

3. Bought-in and mechanical items

3. The 12 modules (bought-in or built in-house)

Temperature (10k NTC with 10k divider), light (LDR with divider), sound level (electret microphone module), magnetic field (linear Hall sensor), soil moisture (capacitive), voltage probe x2 (resistor divider with input protection, low voltage only), IR reflective, water conductivity (AC-driven pair), ultrasonic distance (HC-SR04 class), photogate x2 (U-slot optical sensors, output LOW when the beam is interrupted), button + LED reaction pad. Each module plugs into a 3-pin JST-PH socket: +5 V, signal, GND. Procure from a module supplier with lot test data, or build them on small boards using the same procedure (STANDARD_PROCEDURES A-F).

4. Confidence (read before you order)

- **Order confidence: High**: all parts live-checked, none flagged.
- **Works-as-intended confidence: Medium-High**: Educational kit with a straightforward MCU board; 40 host checks on the statistics and command code.
- **First thing to do:** Run the kit's own experiments on the first unit.
- Rubric and the reasons for every project: `costing/CONFIDENCE.md`. These are judgements, not measurements: nothing here has been built.

5. Where to buy and whom to approach

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Printed circuit board and SMD assembly	JLCPCB is the practical choice for 5 to 100 boards; local PCB shops quote per project (ask for the Gerber ZIP price)	**JLCPCB** (jlcpcb.com), PCBWay (pcbway.com)
Connectors, headers, passives, ICs by LCSC number	element14 India, Mouser India, DigiKey India (genuine, GST invoice, higher price); local electronics markets for common parts	**LCSC** (lcsc.com) with the same cart as JLCPCB
Modules: ESP32 dev boards, SIM7080G, LoRa (RA-08 / RAK3172), pH / EC / DO modules, relay modules, load cells, ultrasonic sensors	Robu.in, Robocraze, ThinkRobotics, Sunrom Electronics, Amazon.in	AliExpress, Seeed Studio, Waveshare
Motors, stepper drivers, wheels	Robu.in, Robocraze, local robotics shops	StepperOnline (omc-stepperonline.com)
Li-ion / LiFePO4 cells and packs, solar panels, MPPT controllers	Authorised battery and solar dealers, Amazon.in (buy protected cells with an invoice)	not recommended by post (shipping rules for batteries)
Laser-cut and 3D-printed parts	Local laser-cutting and 3D-printing shops (search IndiaMART, Justdial or maker spaces): send the SVG/DXF/STL from the repository and ask for a quote per piece	JLC3DP (jlc3dp.com), PCBWay 3D printing
Fasteners, tubing, hardware, enclosures	Local hardware, Robu.in, Amazon.in	n/a

How to approach a vendor: send the project name, the exact file (BOM, CPL, Gerber ZIP, STL or SVG), the quantity, and ask for the price, the lead time, the GST invoice and the warranty or return terms. Keep the reply for your records.